

Solution Requirements

Date	29 September 2026
Team ID	UNM123APEXLAB
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

ComicCraft is a web-based application (FastAPI backend) that uses Google Gemini AI models to generate personalized comic stories and illustrations from user prompts. The requirements below define what the solution will do and how well it will perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High /Medium/Low)
1	Authentication	Users register and log in with email and password (passwords hashed with bcrypt) or Google Sign-In (OAuth 2.0). A session token (JWT) is issued after login and expires automatically after inactivity. A guest mode allows a limited number of free comic generations.	High
2	Authorization levels	Guest: can generate a few sample comics, no saving. Registered User: generate, edit, save, download and delete own comics and view own history. Admin: manage users, monitor usage and API quota, review flagged content and remove inappropriate comics. Users can only access their own data.	High
3	External interfaces	Google Gemini text model API to generate the story script, characters, panel-wise dialogues and scene descriptions. Gemini image model API to generate illustrations for every panel. Google OAuth for login. Database (SQLite/PostgreSQL) to store users and comics, and cloud/local storage for generated images.	High
4	Transactions processing	The user submits a prompt (theme, characters, genre, art style, number of panels). FastAPI validates the input and sends it to Gemini to create the story, then generates one image per panel. Text and images are combined into a comic layout and returned to the user. Each request is logged with a status (pending, success, failed). Failed API calls are retried up to 3 times and a clear error message is shown if they still fail.	High
5	Reporting	User dashboard showing comic history, creation date and status. Download of the finished comic as PDF or PNG. Admin dashboard with total users, number of comics generated, API usage and failure rate.	Medium
6	Business rules, etc.	Each comic has between 4 and 12 panels. Prompts are limited to 500 characters. Daily generation limit per user (for example 5 for guests and 20 for registered users) to control API cost. Unsafe or inappropriate prompts are rejected by content filters. Users can regenerate a single panel without recreating the whole comic. Character names and art style stay consistent across all panels.	High

S.No	Requirement Category	Requirement Description	Priority (High /Medium/Low)
7	Compliance to laws or regulations	Follows data privacy principles (GDPR / India DPDP Act): only minimum data is collected, users can delete their account and comics, and consent is taken before storing data. Follows Google Gemini API terms of use and Google safety policies. AI-generated content is labelled as AI-generated and copyrighted characters are not intentionally reproduced.	Medium
8	Other	Multi-language prompt support (English and Tamil). Option to edit dialogues and captions on the generated comic. Share a comic through a public link. Choice of art styles (manga, cartoon, superhero, watercolor).	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Web pages load quickly. Story text is generated fast and panel images are generated in parallel. A progress indicator is shown during generation so the user is never left waiting without feedback.	Page loads in < 2 seconds. Story script generated in < 10 seconds. Full comic (6 panels) generated in < 60 seconds.
2	Scalability	FastAPI async endpoints and a task queue handle many requests at once. The application is stateless so it can be scaled horizontally, and generated images are stored in external storage.	Supports up to 500 concurrent users. Can scale to 5,000 users by adding server instances without redesign.
3	Security & Data Privacy	Passwords hashed with bcrypt, all traffic over HTTPS/TLS. API keys are kept in environment variables and never exposed to the frontend. Input validation and rate limiting protect against SQL injection, XSS and abuse. Regular dependency updates fix known vulnerabilities.	HTTPS on all endpoints. AES-256 encryption for stored user data. No API key visible in client code. Rate limit of 60 requests per minute per user.
4	Reliability & Availability	Retry and fallback handling for Gemini API failures. Database backups and graceful error messages. Health-check endpoint for monitoring.	99% uptime per month. Automatic retry (3 times) for failed API calls. Daily backups with data recovery within 1 hour.
5	Usability & Accessibility	Simple, clean interface with a prompt form, live preview and easy download. Responsive design for mobile, tablet and desktop. Alt-text for images and keyboard navigation support.	A new user creates the first comic in < 3 clicks. Complies with WCAG 2.1 AA standards. Works on Chrome, Edge, Firefox and Safari.
6	Other	Maintainability: modular code (separate routes, services, models) with documentation and version control on GitHub. Portability: can be deployed on any cloud or local server using Docker.	At least 70% code documented. Deployment on a new server in < 30 minutes.